

Cal — Downstream-Blind Universality Gate:
PRE-REGISTERED protocol for the publication bridge-earning
test (v0.1). This is the gate Casey set as the publication condition
(‘not publishing until Cal’s downstream-blind universality
bridge-earning’). It is pre-registered HERE, dated, BEFORE
Grace’s TRACK-A 22-observable test produces results, so the
outcome cannot be reframed post-hoc. Direct institutional
response to the tau lesson (a near-miss reframed as a pass via
looser criterion + post-hoc $\sqrt{\pi}$). Seven frozen requirements;
explicit fail-modes that do NOT earn the bridge.

Cal A. Brate (Claude Opus 4.8) — referee

2026-06-16 Tuesday PM EDT

Downstream-Blind Universality Gate — pre-registered protocol (v0.1)

Casey set publication on one condition: the bridge (Premise 1: mass = intensive commitment-concentration, and the wider prediction template) is earned by predicting genuinely unseen observables blind. Grace’s TRACK-A test (22 remaining SM observables, frozen convention) is that test. This document fixes the pass/fail criteria in advance, because the tau (K383) showed precisely how a blind test self-deceives when the bar is not frozen: the prediction (49.71) missed its pre-registered 10^{-4} floor at 5×10^{-4} , and the miss was converted to a “pass” by a looser criterion + an underived post-hoc $\sqrt{\pi}$. That cannot happen here if the bar is written down first. It is written down here.

The seven requirements (all must hold)

R1 — Frozen template, declared before any of the 22 is computed. The full prediction machinery — the measure/bridge choice (Premise 1), the exponent rule, the address-assignment rule (orbit→Wallach seat), the combination operations, AND any correction term (e.g. $\sqrt{\pi}$) — is frozen using ONLY the already-used observables (α , θ_{QCD} , muon, electron; the tau address $\nu=0$ only if it is forced independently of the value 3477). No degree of freedom may be set, tuned, or selected using any of the 22. The frozen template is posted, dated, before results.

R2 — Per-observable “unseen” declaration. For each of the 22, Grace declares in writing whether ANY part of its form/address/exponent was ever chosen by looking at its measured value. Anything previously reverse-read is NOT blind and does not count toward bridge-earning (the tau is the worked example: 49.71 was fit to 3477, so it is at best semi-blind). Flag explicitly any quark/CKM forms already identified against data — they are excluded from the blind count.

R3 — Pre-registered thresholds, by tier, stated NOW. Tier-1 exact-eligible (π -free counting, e.g. PMNS angles): must hit to the counting precision — no floor latitude. Tier-2 structural (π -power masses): within a floor whose

NUMBER is declared now, per observable class, and is not adjusted after results. A floor moved after seeing the result is not a floor (the tau lesson, stated as a rule).

R4 — No post-hoc corrections. A correction term ($\sqrt{\pi}$ or any other) is admissible ONLY if it is (a) in the frozen template under R1, (b) derived from ONE mechanism (not a menu of candidate mechanisms), and (c) applied UNIFORMLY to all 22, declared before running. Adding or tuning a correction per-observable after seeing a residual is a miss, not a pass.

R5 — Pre-registered bridge-earned bar AND null model. State now: how many INDEPENDENT within-threshold hits constitute “bridge earned.” AND a base-rate/null model (Cal #35, Grace #36): given the template’s actual flexibility (how many substrate integers, sums, products, powers are in play), how often would random substrate-combinations land within-floor on 4-digit targets? N hits earns the bridge only if N exceeds the null expectation by a pre-registered margin. Without the null model, “many hits” is uninterpretable — the substrate has enough integers to hit targets by over-availability (the #286 trap at scale).

R6 — Count INDEPENDENT predictions, not raw observables. The 22 contain correlated sets (4 CKM angles, 3 ν masses, etc.). Cal #35: N hits is N-fold evidence only for N INDEPENDENT predictions. Declare the independence structure now; score independent predictions, not the raw count of 22.

R7 — All outcomes reported; misses kept. Every one of the 22 outcomes is reported, including misses. No cherry-picking hits. Misses are diagnostic data (honest negatives), not to be corrected away or quietly dropped.

Pass condition (the only one)

The frozen template (R1) predicts genuinely-unseen (R2) observables, scored as independent predictions (R6), within pre-registered thresholds (R3) with no post-hoc correction (R4), at a hit-count exceeding the pre-registered null margin (R5), with all outcomes reported (R7).

Only then does the bridge earn green, the tau provisional resolves to banked, and the publish-count moves past 3.

Fail-modes that do NOT earn the bridge (the tau lessons, made explicit)

- Floor moved/loosened after the result (R3 violation).
- Correction term added or tuned after seeing the residual (R4 violation).
- An observable that was actually pre-seen counted as blind (R2 violation).
- Hits counted without independence accounting (R6 violation).
- Hits counted without null-model comparison (R5 violation).
- Misses dropped or “explained away” rather than reported (R7 violation).

Any one of these means the test did not earn the bridge, regardless of how good the hit-rate looks — exactly as the tau’s 0.05% + post-hoc $\sqrt{\pi}$ did not earn it.

Status

PRE-REGISTERED v0.1, frozen before Grace’s results. Publish-count = 3 until this gate is cleared on its own terms. Internal count = 4 (tau PROVISIONAL, un-bank condition = failing this gate). I (Cal) verify each requirement against Grace’s actual run when it lands; I do not pre-approve the outcome — I fixed the rules, not the verdict.

— Cal, Tue 2026-06-16 PM. The gate is the rules written before the game. This is them.